

Ernest Bennin

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PROFILE

An ABD Ph.D. candidate specializing in ligand design for enantioselective catalysis and chemical sensing. Developed innovative tethered ligands for enhanced dirhodium complexes, resulting in published research. Demonstrated expertise in chemical synthesis, spectroscopy, and analytical instrumentation, driving impactful solutions to detection challenges. Proven ability to mentor and collaborate within interdisciplinary teams, fostering a culture of innovation and continuous improvement. Education includes a Doctor of Philosophy in Analytical Chemistry from the University of Tennessee, Knoxville, expected in May 2027, and a Bachelor of Science in Chemistry from the Kwame Nkrumah University of Science and Technology.

SKILLSET

- Organic reaction setups, Inorganic reaction setups, Crystal growth, Schlenk and glovebox techniques, manual and automated flash column chromatography, crystallization, recrystallization, and HPLC setup and analysis.
- Ultraviolet-Visible Spectroscopy, Nuclear Magnetic Resonance (NMR (^1H , ^{13}C , ^{19}F , ^{31}P)), Fluorescence spectroscopy, Infrared (IR) spectroscopy, Circular dichroism.
- Microsoft Office, Gaussian, Data analysis, MestReNova, ChemDraw.
- Leadership, Written and Oral Communication, Problem-Solving, Teamwork, Organization, Project Management, Creativity, and Adaptability.

RECOGNITIONS

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| • NOBCChE Advancing Science Conference Grant (ASCG), Tier 1 | Jul. 2026 |
| • NSF-Center for Computer-Assisted Synthesis Professional Development Award | Apr. 2026 |
| • NOBCChE Leadership Academy, 2026 Cohort | Apr. 2026 |
| • NSF-Center for Computer-Assisted Synthesis Professional Development Award | Aug. 2025 |
| • NOBCChE Advancing Science Conference Grant (ASCG), Tier 2 | Jun. 2025 |
| • NSF-Center for Computer-Assisted Synthesis Professional Development Award | May 2025 |
| • The Graduate Student Senate Travel Award | Oct. 2024 |
| • NSF Center for Computer-Assisted Synthesis Research Grant | Jan 2023-Aug. 2025 |
| • FY23 Grad School Fellowship | Aug. 2022-present |
| • Graduate Teaching/Research Assistantship Award | Aug. 2022-present |
| • GETFUND Scholarship, Ghana (Awarded to a brilliant student) | Aug. 2017-Nov 2021 |

WORK AND RESEARCH EXPERIENCE

Chemistry Graduate Research Assistant

Nov. 2023- Present

The University of Tennessee, Knoxville

- Led research on strategic ligand design in modulating the axial site on dirhodium paddlewheel complexes, towards enhanced asymmetric cyclopropanation, resulting in a **first-author** publication in Dalton Transactions
- Conducted key experiments on chromogenic detection of the organophosphate nerve agent simulant DCP using rhodium(II,II) paddlewheel complexes, contributing to publications in ACS Sensors and ACS Omega.
- Designed and synthesized a tethered ligand to modify dirhodium paddlewheel complexes for improved axial coordination and enantioselectivity in diazo-mediated cyclopropanation reactions.
- Synthesized dirhodium paddlewheel complexes to explore new sensing approaches for organophosphorus nerve agent simulant (DCP), resulting in one published and one submitted manuscript.
- Collaborated with interdisciplinary teams, including NSF-CCAS, CCDC-CBC, to solve detection challenges and advance sensor and catalyst development.
- Mentored undergraduate students in laboratory techniques and scientific communication, conducted lab safety checks and waste management, and coordinated academic and team-building activities to support a collaborative research environment.

Undergraduate Research Assistant

May 2021- Aug 2022

Department of Chemistry, Kwame Nkrumah University of Science and Technology

- Conducted groundwater quality analysis to assess contamination levels and public health risks in the Offinso-North District, which informed recommendations for safer water use.
- Collaborated on heavy metal pollution studies in soil from the Oti-Dompoase dumpsite to evaluate environmental impact and support regional pollution mitigation efforts, which led to one publication.

Assisted licensed pharmacists as an intern

May 2018-Aug 2018 / May 2019-Aug 2019

Offinso-North District Government Hospital, Ashanti region-Ghana

Supervisor: Dr. Sampany Kese

- Prepared and dispensed prescription medication under supervision, ensuring patient safety and dosage accuracy
- Packaged and labelled medications, formulated specialized dosage forms, and maintained clear instructions to support compliance and reduce errors
- Assisted patients with inquiries and transactions while managing pharmacy software, inventory, and records to improve service and maintain regulatory compliance.

TEACHING EXPERIENCE

Head of Teaching Assistants

May 2023- Jul 2023

Department of Chemistry - The University of Tennessee, Knoxville

Coordinator: Dr. Nahla Abu Hatab

- Supervised and supported a team of teaching assistants during laboratory sessions (Foundation of Organic Chemistry Lab, CHEM 269) to ensure accurate demonstration and student comprehension.
- Assisted students and teaching assistants with technical questions to promote lab safety and success.
- Monitored in-person examinations and graded assessments to maintain academic integrity and timely feedback.

Chemistry Teaching Assistant

^a*The University of Tennessee*

Aug 2025- present

^a*The University of Tennessee*

Jan 2023- May 2023

^b*Kwame Nkrumah University of Science and Technology*

May 2021- Aug 2022

Coordinators: Dr. Carroll Xian^a, Dr. Nahla Abu Hata^a, Dr. Osei Akoto^b

- Demonstrated chemistry labs (Analytical Chemistry V Lab, CHEM 369, Analytical Chemistry VI Lab, CHEM 370, General Chemistry I Lab, CHEM 123, and Foundation of Organic Chemistry Lab, CHEM 269) to undergraduate students, helping connect theory with practice for a deeper understanding.
- Graded reports and examinations to provide students with constructive feedback and insights into performance.
- Organized tutorials and problem-solving sessions to reinforce lecture content and improve exam preparation.
- Supported lecturers with instructional duties, enhancing the delivery and effectiveness of course content (Instrumental Methods of Analysis II, CHEM 461, and Environmental Chemistry, CHEM 462).

SELECTED PUBLICATIONS

1. **Ernest Bennin**, Ryan O'Connell, Cristian Zavala, Gisela G. Montiel, Olaf Wiest*, and Ampofo Darko*. Strategic Ligand Design for Modulating Axial Sites in Dirhodium Catalysts: Toward Enhanced Asymmetric Cyclopropanation. Published, *Dalton Trans.* Jul 30, 2026. [ID DT-ART-05-2026-001202.R1](#)
2. Desiree Moore, Eric D. Fussell, **Ernest Bennin**, Bukola Ogunyemi, Neal D. Kline*, and Ampofo Darko*. Selective Sensing of Nerve Agent Simulant via Dirhodium(II, II) Paddlewheel Complexes Featuring Pendant Hemilabile Thioether Ligands. *ACS Omega*. Published online, May 14, 2026.
3. Ernest Bennin. "[How federal research funding transformed my path.](#)" *Knox news*, Jan. 15, 2026. (Op-Ed)
4. Eric D. Fussell, **Ernest Bennin**, Neal D. Kline*, Sarah S. Hirshbeck, and Ampofo Darko* Chromogenic detection of the organophosphate nerve agent simulant DCP mediated by rhodium(II, II)paddlewheel complexes. *ACS Sensors*. 2024, 9(5), 2325-2333

5. Nkansah, M. A., Kofi Adzibolosu, A. S., Kumah, M. E., **Bennin, E.**, Adrewie, D. Heavy Metal Pollution Studies of Soil from the Oti-Dompoase Dumpsite in Kumasi. *Fine Chemical Engineering*. **2024**, 5, 397-413
6. **Ernest Bennin**, Gisela G. Montiel, Olaf Wiest*, and Ampofo Darko*. A New Ligand Design Based on Halogen Interactions Enables Chiral TACLs Dirhodium Paddlewheel Catalysts to Achieve Enhanced Enantioselectivity in C-H Insertion and Cyclopropanation—Manuscript in Progress.

SELECTED PRESENTATIONS

1. **Oral Presentation**, Axially Coordinated Chiral Ligands on Dirhodium(II,II) Complexes for Enhanced Asymmetric Cyclopropanation. NSF Center for Computer Assisted Synthesis (C-CAS) Annual meeting. Notre Dame, South Bend, IN. May 21, 2026.
2. **Poster Presentation**, Chromogenic Detection of the Organophosphorus Nerve Agent Simulant DCP Mediated by Rhodium(II, II) Paddlewheel Complexes. National Organization for the Professional Advancement of Black Chemists and Chemical Engineers (NOBCChE) conference. Atlanta, GA. Sep. 2025.
3. **Poster Presentation**, Asymmetric cyclopropanation on dirhodium complexes with tethered axially coordinated chiral ligands. (a) American Chemical Society, Division of Organic Chemistry's National Organic Symposium, Troy, NY. Jun. 2025. (b) National Organization for the Professional Advancement of Black Chemists and Chemical Engineers (NOBCChE) conference. Orlando, FL. Sep. 2024.
4. **Poster Presentation**, DFT modelling of dirhodium complexes with tethered axially coordinated chiral ligands for asymmetric cyclopropanation. NSF-Center for Computational and Assisted Synthesis. UC-Davis, LA. Jun. 2024.
5. **Oral Presentation**, Asymmetric cyclopropanation on dirhodium complexes with tethered axially coordinated chiral ligands. (a) 3rd Year seminar, The University of Tennessee, Knoxville, TN, May 2025 (b) Candidacy examination, The University of Tennessee, Knoxville, TN, Mar. 2024.
6. **Poster Presentation**, Determination of physicochemical parameters, nutrients, heavy metal content, and human health risk assessment of groundwater from the Offinso-North District of Ashanti Region, Ghana. (a) Undergraduate poster session, Department of Chemistry, KNUST, Kumasi, Ghana 2021 (b) Ghana Science Association, College of Science, KNUST, Kumasi, Ghana 2021.
7. **Oral Presentation**, Determination of physicochemical parameters, nutrients, heavy metal content, and human health risk assessment of groundwater from the Offinso-North District of Ashanti Region, Ghana. Undergraduate project defense, Department of Chemistry, KNUST, Kumasi, Ghana, 2021.

LEADERSHIP ROLES

- ASCG committee member** for NOBCChE conference planning, Cincinnati, OH Feb 2026
- Reviewed and evaluated abstracts to select ASCG grant recipients for conference attendance
- President-** NOBCChE Chapter, University of Tennessee Mar 2025-Present
- Led the organization on a tour of the Neutron Scattering Facility at Oak Ridge National Laboratory, contributing to the group being recognized as UTK's Best Academic Organization for the first time under my leadership.
 - Led the organization to earn the GSS-UTK Outstanding Graduate Student Organization Award for outstanding contributions to graduate student life and campus engagement Apr. 2026
- President** - Ghana Students' Association, University of Tennessee, (GSA-UTK) Aug 2025-Present
- Collaborated with team members to foster an inclusive community for Ghanaian students by organizing cultural and social events such as the Meet and Greet and Ghana Independence Day celebration to promote belonging and connection.
- Deputy Safety Officer-** Darko Research Group, University of Tennessee. Aug 2023-May 2024